

PERIPHERAL VASCULAR DISEASE

Original Studies

Endovascular Therapy for Angio-Seal™-Related Acute Limb Ischemia: Perioperative and Long-Term Results

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Objectives: To investigate the perioperative and long-term outcomes of endovascular therapy for Angio-seal™-related acute limb ischemia. **Background:** Currently, limited data are available on the optimal treatment strategy for Angio-seal™-related acute lower limb ischemia. **Methods:** The prospectively maintained patient database of our institution was retrospectively searched to identify all patients who developed acute lower limb ischemia after use of the Angio-Seal™ and received endovascular treatment from January 2010 to February 2016. The clinical and follow-up data were evaluated. **Results:** Thirty-two patients with Angio-Seal™-related acute limb ischemia underwent endovascular therapy, resulting in an approximated incidence of 0.29% of all implanted devices. The overall procedural success rate was 96.9%. With regard to the patients who underwent successful endovascular treatment, the culprit lesion was located in the common femoral artery in 22 (71.0%) cases, the femoral artery bifurcation in 3 (9.7%) cases and the superficial femoral artery in 6 (19.4%) cases. Ten (31.3%) patients had thrombosis in other distal lower limb arteries ipsilateral to Angio-seal™ use. Eighteen (58.1%) patients underwent balloon angioplasty alone, while nine (29.0%) patients underwent balloon angioplasty and thrombolysis, and four (12.9%) patients underwent stent implantation. One patient suffered from minor bleeding at the site of application of the Angio-seal™. During an average of 43.5 ± 22.9 months of follow-up, three patients with symptomatic restenosis underwent a second successful balloon angioplasty and remained asymptomatic until the last follow-up. **Conclusions:** Balloon angioplasty with selective thrombolysis or stent placement was safe and effective, with a low incidence of complication and restenosis. © 2017 Wiley Periodicals, Inc.

Key words: balloon angioplasty; stents; acute limb ischemia; Angio-seal; vascular closure devices

INTRODUCTION

Vascular closure devices (VCDs) are widely used to achieve hemostasis after cardiac and peripheral vascular interventions via femoral access. Although randomized controlled trials and meta-analyses have shown no significant difference in bleeding complications from VCDs compared with manual compression, the utility of VCDs could shorten the time to hemostasis and ambulation, reduce hospital stays, and even decrease the incidence of short-term mortality [1–4]. Notwithstanding these potential advantages, the use of VCDs can result in adverse vascular events, including stenosis, hematoma, bleeding, arteriovenous fistula, pseudoaneurysm, thrombosis, and infection [2,5].

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TABLE I. Reported Therapy for Angio-seal™-Related Acute Limb Ischemia ($n \geq 5$)

Author (Reference)	Year	Patients (<i>n</i>)	Localization	Therapy	Follow-up (months)
Goyen et al. [18]	2000	5	CFA, SFA, DFA, PA	4 Endovascular, 1 Surgery	0–6
Steinkamp et al. [20]	2001	16	CFA, SFA, DFA	13 Endovascular, 3 Surgery	0–6
Kirchhof et al. [14]	2002	10	CFA, SFA	Surgery (most)	2–3
Thalhammer et al. [15]	2004	14	CFA, SFA, DFA, EIA, CIA	Surgery (most)	n.a.
Siani et al. [11]	2006	5	CFA, SFA, PA	Surgery	16 (mean)
Wille et al. [12]	2006	6	CFA, SFA,	Surgery	n.a.
Kadner et al. [16]	2008	7	CFA	Surgery	1–12
Pontón et al. [9]	2009	7	CFA, SFA	Surgery	5–11
Thalhammer et al. [6]	2010	15	CFA, SFA	Endovascular	4–79
Georg et al. [13]	2011	5	CFA, SFA, EIA, HA	Surgery	n.a.
Siani et al. [10]	2011	8	CFA, SFA, PA	Surgery	36 (mean)
Klocker et al. [8]	2012	38	CFA, SFA, IA	Surgery	3.6–67.2
Rekik et al. [21]	2013	12	CFA	Endovascular	32 (mean)
Amer et al. [7]	2015	43	CFA, SFA, FAB, PA/BKA	16 Endovascular, 27 Surgery	n.a.
Ganai et al. [19]	2016	12	CFA, EIA, PA	9 Endovascular, 3 Surgery	2–53

CFA, common femoral artery; SFA, superficial femoral artery; DFA, deep femoral artery; PA, popliteal artery; EIA, external iliac artery; CIA, common iliac artery; HA, humeral artery; IA, iliac artery; FAB, femoral artery bifurcation; BKA, below knee artery; n.a., not available.

Angio-Seal™ (St. Jude Medical, The United States) is one of the most commonly used VCDs, which deposits a small collagen plug into the puncture channel to mechanically seal the puncture defect [6,7]. The most serious complication related to the application of the Angio-Seal™ closure device is acute limb ischemia due to femoral artery stenosis and occlusion, which requires revascularization treatment primarily to restore blood flow through lower limb arteries and to relieve ischemic symptoms. Currently, optimal revascularization strategy for Angio-seal™-related acute lower limb ischemia remains unknown [7]. Various small-scale studies have reported successful surgical procedures, including patch angioplasty, interposition graft, thrombectomy, endarterectomy, etc. [7–16] (Table I). However, data on endovascular therapy is limited to small sample sizes and short-term follow-up periods [6,7,17–21] (Table I). Therefore, the aim of this study was to investigate the perioperative and long-term outcomes of endovascular therapy for Angio-seal™-related acute limb ischemia.

MATERIALS AND METHODS

Patient Population

The prospectively maintained patient database of our institution was retrospectively searched to identify all patients who developed acute lower limb ischemia after use of the Angio-Seal™ (St. Jude Medical, The United States) and received endovascular treatment from January 2010 to February 2016. Patients were considered for endovascular treatment if they had clinical symptoms of moderate/severe claudication or ischemic rest pain and femoral artery stenosis greater than 70% secondary to Angio-Seal application that could not be

resolved by conservative treatment. At our institution, surgical treatment was only performed unless endovascular therapy was unsuccessful. Before the procedure, all patients provided written informed consent.

Interventional Technique and Follow-Up

The interventional procedures were performed under local anesthesia through contralateral femoral artery or right brachial artery access. Using the Seldinger technique, a 6-French introducer sheath was positioned, and heparin (100 U/kg) was administered. A 5-French Cobra 2 (Terumo, Japan) or MPA1 (Cordis, The United States) diagnostic catheter was introduced into the external iliac artery over a 0.035-inch HiWire angled hydrophilic guidewire (Cook, The United States). Selective lower limb angiography was first performed to determine the location and characteristics of the lesion. Then, a 6-French 90 cm shuttle flexor introducer long sheath (Cook, The United States) was positioned near the lesion. The majority of stenotic and occlusive lesions could be crossed with a 0.035-inch HiWire angled hydrophilic guidewire, while some lesions required other types of guidewire, including 0.018-inch V-18 steerable guidewire (Boston Scientific, The United States) or 0.035-inch Work Amplatz (PBN Medicals, Denmark). All lesions were dilated using graded balloons first with a long inflation time up to 2 min and repeated dilations as needed. With regard to patients with thrombosis in ipsilateral lower limb arteries, 200,000–500,000 units urokinase was infused through the long sheath. If flow-limited dissection and/or residual stenosis greater than 50% was revealed by angiogram, a self-expandable stent (Biotronik, Germany) was implanted. After the procedure, low-molecular weight heparin (80–100 U/kg,

TABLE II. Baseline Clinical Characteristics (n = 32)

Variables	n (%)
Age (years)	62.5 ± 10.0
Male	22 (68.8)
Hypertension	26 (81.3)
Ongoing smoking	12 (37.5)
Hyperlipidemia	19 (59.4)
Diabetes mellitus	15 (46.9)
Body mass index (kg/m ²)	27.1 ± 4.3
Obesity	10 (31.3)
Renal insufficiency ^a	12 (37.5)
LVEF < 40%	9 (28.1)
Coronary artery disease	30 (93.8)
Peripheral artery disease	15 (46.9)
Sheath size	
6 French	28 (87.5)
7 French	3 (9.4)
8 French	1 (3.1)
History of same site puncture	22 (68.8)
History of using Angio-seal at same site	9 (28.1)

LVEF, left ventricular ejection fraction.

^aBaseline serum creatinine ≥ 1.5 mg/dL.

Q12H) was administered for patients with thrombosis. Aspirin and clopidogrel were continued for 1 month for those who underwent balloon angioplasty and for 3 months for those that received a stent.

Telephone or clinic interviews were conducted at 1 month, 6 months, and every 6 months after the procedure by a dedicated research coordinator. Symptoms of lower limb ischemia and ankle brachial index were examined, and complications (death, stroke, myocardial infarction, etc.) were recorded. Lower limb computerized tomographic or transcatheter angiography was performed if any patient had symptomatic lower limb ischemia ipsilateral to Angio-seal™ use during follow up.

Statistical Analysis

Continuous variables were reported as the mean ± standard deviation, while categorical data were given as counts and percentages. Differences between angiographic stenosis measured before and after the procedure were tested by paired *t*-tests. All analyses were performed using the software SPSS 17.0 (SPSS Inc.). All probability values were 2-sided, and a *P* value < 0.05 was considered to indicate statistical significance.

RESULTS

Baseline Clinical Characteristics

Thirty-two patients underwent endovascular therapy for Angio-seal™-related acute limb ischemia between January 2010 and February 2016. Over this period, an estimated 10,700 Angio-seal™ closure devices were used at our institution, resulting in an approximated incidence of 0.29%.

TABLE III. Procedure Characteristics of the 31 Lesions that Underwent Successful Endovascular Therapy

Variables	n (%)
Puncture site	
Contralateral femoral artery	24 (77.4)
Right brachial artery	7 (22.6)
Lesion location	
Common femoral artery	22 (71.0)
Femoral artery bifurcation	3 (9.7)
Superficial femoral artery	6 (19.4)
Lesion side	
Right	27 (87.1)
Left	4 (12.9)
Stenosis of pre-procedure	97.5 ± 5.1
Stenosis of post-procedure	23.9 ± 14.5
Thrombosis in ipsilateral distal lower limb arteries	10 (32.3)
Treatment strategy	
Balloon angioplasty	18 (58.1)
Balloon angioplasty + thrombolysis	9 (29.0)
Balloon angioplasty + Stent	4 (12.9)
Dissection after balloon angioplasty	10 (32.3)

Patient ages ranged from 26 to 76 (mean 62.5 ± 10.0) years. Twenty-two (68.8%) patients were male, and the other ten (31.2%) were female. Except for one female with Takayasu's arteritis, the etiology of all other patients was atherosclerosis. Twelve (37.5%) patients had renal insufficiency and nine (28.1%) had low left ventricular ejection fraction (<40%) measured by echocardiography. In addition, 30 (93.8%) patients suffered from coronary artery disease, and 15 (46.9%) had peripheral artery disease. 5 (15.6%), 16 (50.0%), and 11 (34.4%) patients suffered from moderate, severe claudication and ischemic rest pain, respectively. The index procedures corresponded to coronary interventions in 29 cases and peripheral vascular interventions in 3 cases. Of them, 28 interventional procedures were performed using 6-French introducers, while 3 and 1 interventional procedures were performed using 7- and 8-French introducers. Moreover, 22 (68.8%) patients had a history of same-site puncture, and 9 (28.1%) had previous Angio-seal™ applications at same site. The interval between the onset of acute limb ischemia and the interventional procedure was 0–25 (mean 5.2 ± 6.4) days. The patients' baseline clinical characteristics are summarized in Table II.

Interventional-Related Data

The overall procedural success rate was 96.9% (32/33). One patient with a previous left superficial femoral artery occlusion and an ipsilateral common femoral artery occlusion following Angio-seal™ use failed to undergo endovascular treatment because the guidewire could not pass through the lesion. The patient underwent iliofemoral bypass successfully. The procedure characteristics of 31 patients with lesions underwent successful

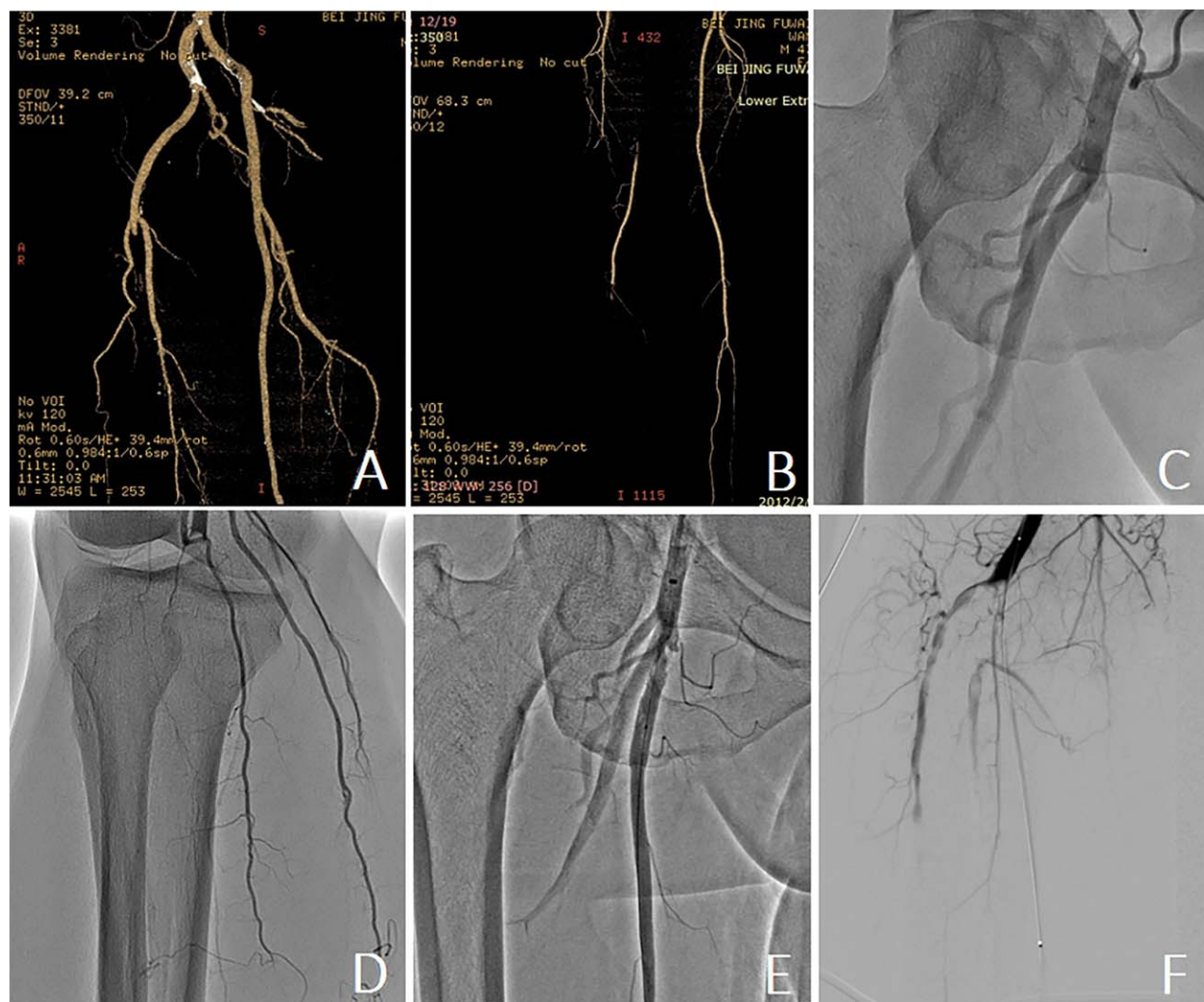


Fig. 1. Representative angiography of balloon angioplasty and thrombolysis (A) and (B) Computerized tomographic angiography revealed right superficial femoral artery occlusion and below-knee artery thrombosis. (C) and (D) Selective lower limb angiogram demonstrated the findings of the computerized tomographic angiography. (E) and (F) Lower limb angiogram revealed a 50% residual stenosis in the proximal right

superficial femoral artery and partial recanalization in the below-knee artery after balloon angioplasty (Passeo-35 5.0 × 40 mm and Passeo-35 6.0 × 40 mm balloon in superficial femoral artery and Passeo-18 3.5 × 120 mm balloon in below-knee artery) and thrombolysis (transcatheter infusion of 500,000 unit urokinase).

endovascular therapy are shown in Table III. The puncture site in 24 (77.4%) patients was the contralateral femoral artery, while in the other seven (22.6%) was the right brachial artery. The culprit lesion was located in the common femoral artery in 22 (71.0%) cases, the femoral artery bifurcation in 3 (9.7%) cases and the superficial femoral artery in 6 (19.4) cases. Of them, 22 lesions were occlusive, while the other 9 lesions were stenotic. Ten (32.3%) patients had thrombosis in the distal lower limb arteries ipsilateral to Angio-seal™ use. All lesions first underwent balloon angioplasty using angioplasty balloons with maximal diameters of 5–7 (6.3 ± 0.7) mm.

Eighteen (58.1%) patients underwent balloon angioplasty alone and nine (29.0%) patients underwent balloon angioplasty and thrombolysis (Fig. 1). Ten patients suffered from vessel dissection after balloon angioplasty, of whom four (12.9%) patients underwent stent implantation with stent diameters of 6–7 mm due to flow-limited dissection. The mean degree of angiographic stenosis was reduced from $97.5\% \pm 5.1\%$ to $23.9\% \pm 14.5\%$. One patient suffered from minor bleeding at the site of Angio-seal™ application and successfully received manual compression with no sequela. No other adverse events occurred in the perioperative period.

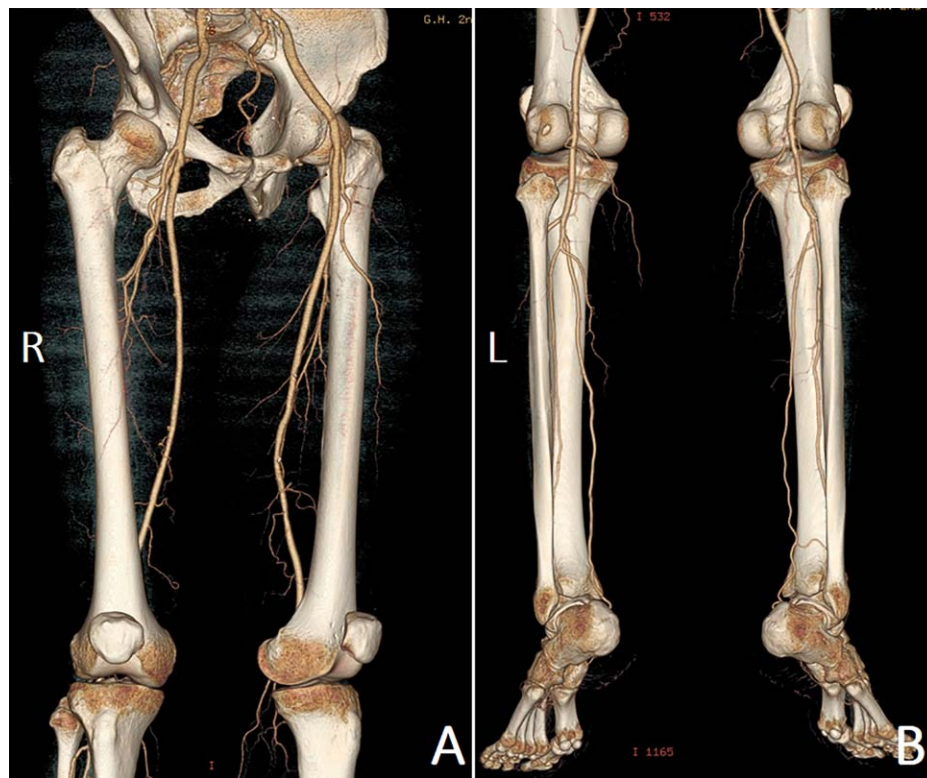


Fig. 2. Representative computerized tomographic angiography demonstrated good blood flow in the right superficial femoral artery (A) and the below-knee artery (B) at 4 months follow-up after balloon angioplasty and thrombolysis. The ankle brachial index of right lower limb increased from 0 before endovascular treatment to 1.14 at the 4-month follow up.

Long-Term Outcome

During an average of 43.5 ± 22.9 months of follow-up (range 6–79 months), 28 patients remained asymptomatic (Fig. 2), and another three (9.7%) patients developed symptomatic restenosis and required reintervention at 4 months, 8 months, and 6 months, respectively; the former two patients were treated by balloon angioplasty alone, and the other patient underwent stent implantation initially. All three patients underwent a second successful balloon angioplasty and remained asymptomatic until the last follow-up. Two patients died unrelated to lower limb ischemia (1 cardiac shock at 29 months and 1 traffic accident at 10 months), one patient suffered from myocardial infarction at 18 months, and one patient had a minor stroke at 23 months. There were no other adverse events during the follow-up period.

DISCUSSION

To the best of our knowledge, the present study represents the largest consecutive series and longest follow-up of patients with acute limb ischemia related

to the use of the Angio-seal™ closure device who underwent subsequent endovascular therapy. The estimated incidence of acute limb ischemia due to femoral artery stenosis and occlusions was 0.29% in our sample. Balloon angioplasty with selective thrombolysis or stent placement was safe and effective, with a high procedural success rate and a low incidence of complication and restenosis during an average of 44 months of follow-up.

The increasing number of annual endovascular procedures worldwide results in the widespread use of VCDs, which is also accompanied by an elevated incidence of related complications. As one of the most severe complications, acute lower limb ischemia has been described after the use of all types of VCDs [1,2,4], with a prevalence of 0.1%–4.2% after use of the Angio-seal™ closure device [6,7,21,22]. However, in the absence of clinical studies with large sample sizes, the optimal treatment strategy for patients with Angio-seal™-related acute limb ischemia remains in dispute. Current treatment approaches for such patients include conservative approach, endovascular treatment, or surgical management, and which approach should

be used depends upon the severity of ischemic symptoms, the anatomical location of the lesions and the patient's individual characteristics [6]. After a patient suffers from moderate/severe claudication or ischemic rest pain, medical therapy alone is not effective, and lower limb artery revascularization is needed. Surgical approaches have been reported in some samples with a high procedural success rate [8–10]. Nevertheless, the administration of dual antiplatelet therapy after previous interventional procedures, high incidences of death and reoperation (4% and 20%, respectively, reported by Klocker et al. [8]), and long hospital stays have precluded the widespread use of reconstructive surgery.

Currently, some case reports and small-sample studies have indicated the feasibility of endovascular therapy. A 45-year-old male patient developed acute ipsilateral lower limb ischemia due to occlusions of the popliteal and crural vessels after receiving an Angio-sealTM, as reported by Schiele et al. The patient was successfully treated by percutaneous thrombectomy, thrombolysis and balloon angioplasty [17]. Thalhammer et al described 15 patients with Angio-sealTM-related acute limb ischemia, of which eleven patients were treated with balloon angioplasty and four (26.7%) with stent implantation. During a median time of 40 months follow up, 2 (13.3%) patients underwent repeat balloon angioplasty due to symptomatic restenosis [6]. Rekik et al. reported 12 patients who developed acute lower limb ischemia after use of the Angio-sealTM [21]. Seven patients underwent balloon angioplasty alone and five (41.7%) underwent stent implantation. Two (16.7%) patients developed symptomatic restenosis and underwent balloon angioplasty and stent placement, respectively. In a recent publication by Amer et al., 43 patients suffered from Angio-sealTM-related acute limb ischemia, and endovascular treatment was performed in 16 (37%) patients, including fifteen patients (94%) who underwent balloon angioplasty, nine patients (56%) stent implantation, eight patients (50%) thrombectomy and three patients (19%) local fibrinolysis. Two patients died due to cardiogenic shock and multiorgan failure, respectively, during the perioperative period [7]. However, due to small sample sizes and short follow-up periods, the clinical outcomes of endovascular approaches must be assessed to systematically investigate larger samples. In our study, 32 patients with Angio-sealTM-related acute limb ischemia underwent endovascular therapy, and the overall procedural success rate was 96.9%. Of the 31 patients who underwent successful endovascular therapy, 32.3% had thrombosis in other distal lower limb arteries ipsilateral to Angio-sealTM use. Eighteen (58.1%) patients underwent balloon angioplasty alone, while nine (29.0%) patients underwent balloon angioplasty and

thrombolysis, and four (12.9%) patients underwent stent implantation. Only one patient developed minor bleeding at the site of Angio-sealTM application. During an average of 44 months of follow-up, three patients (9.7%) with symptomatic restenosis underwent a second successful balloon angioplasty. Our study showed that endovascular treatment was safe and technically feasible for resolving lower limb ischemia for patients with Angio-sealTM-related acute limb ischemia in a high-volume cardiovascular referral center. The proportion of patients who underwent stent placement (12.9%) in our series was lower than in the studies mentioned above (26.7%–56%) [6,7,21]. This decision was based mainly on the following two points: first, “watchful waiting” should be considered after suboptimal balloon angioplasty because the collagen plug dissolves after 60–90 days, the dissection heals with time, the peripheral thrombus disappears after thrombolysis and anticoagulation; second, stent trauma, deformation, and in-stent restenosis due to endometrial hyperplasia are common in stent placement of the infrainguinal arteries.

Study Limitations

Our study has several limitations. First, this was a retrospective study, and all the limitations inherent to this type of study are obvious. Second, transcatheter angiography or CT angiography was not performed routinely during the follow-up period; therefore, the real restenosis rate may be slightly underestimated. Additionally, due to the absence of a control group (e.g., surgical approach), the results can only be compared with data from the literature.

CONCLUSIONS

Balloon angioplasty with selective thrombolysis or stent placement was a safe and effective treatment strategy for patients with Angio-sealTM-related acute limb ischemia, with a low incidence of complication and restenosis during an average of 44 months of follow-up.

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